The Anatomy of Diffusion Models

Pin-Jing, Li

ouo.ee11@nycu.edu.tw
Advisor: Professor Yu-Chih (Jerry) Huang
National Yang Ming Chiao Tung University

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Diffusion Overdose

It's hard to not heard of the name of diffusion models in the quickly expanding generative AI era.

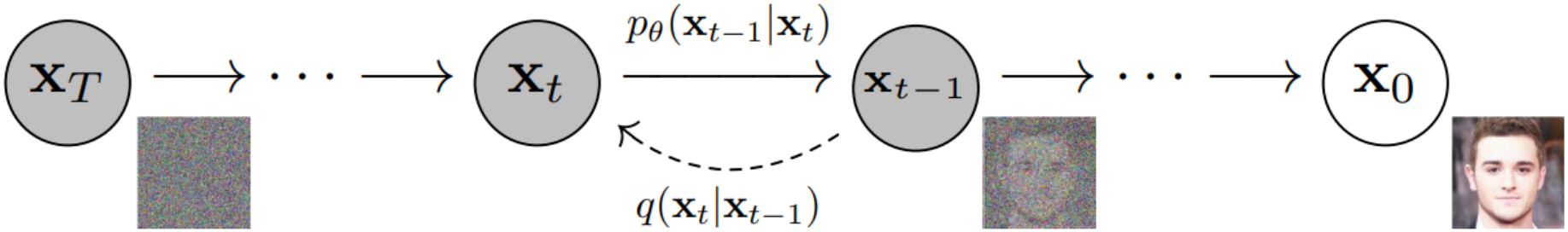


Figure: Diffusion Denoising Probabilistic Model.

Goal of Generative Models

Given training samples $x_0 \sim p_{\text{data}}(x)$, the objective is to train a model such that

$$p_\theta(x_0) \approx p_{\text{data}}(x),$$

where

- p_{data}: the true (unknown) data distribution,
- p_θ: the learned distribution parameterized by the model.

Intuition: learn to generate new samples $\hat{x}_0 \sim p_\theta$ that are indistinguishable from real data.

Intuition of the Diffusion Model Machenism

Training

- Starting from drawing a data from the source distribution $\mathbf{x}_1 \sim p_1$,
- Construct an accessible time-dependent evolution $\{\mathbf{x}_t\}$ from $t = 1 \to 0$,
- Find a way to reverse the evolution.

DDPMs formulate learning by discretizing the forward noising process into sufficiently fine steps, and training the model to denoise an infinitesimal corruption at each timestep.

Inference

- Sample from the known, simple latent,
- Use the trained "inverter" to recover an $\hat{\mathbf{x}}_1$

To better understand the mathematical explanation of DDPM, we start from another classical generative model structure.

Variational Autoencoders (1)

VAE is one of the canonical examples of a generative model. In here our task is,

- for a set of observed data $x \sim p_X$
- VAE defines a parameterized model $p_\theta(x)$ that aims to match the true distribution $p_X(x)$

Directly parameterizing $p_\theta(x)$ in high-dimentional space is not computationally-friendly.

Variational Autoencoders (2)

To get access to the distribution $p(x)$, we introduce the latent space.

Suppose there is a prior of the latent variable with lower dimensionality $z \sim p(z)$, we can now rewrite the learned distribution as if it's induced by marginalizing over the latent space,

$$p(x) = \int p_\theta(x|z)p(z)dz$$

where

- z the latent variable
- $p(z)$ the prior distribution of the latent variable.
- $p_\theta(x|z)$ the parameterized decoder that decodes the latent space variable back into the data.

We can now generate x by sampling z and then decoding with the decoder $p_\theta(x|z)$.
But how to train the decoder model $p_\theta(x|z)$?

Variational Autoencoders (3)

An idea is to use the posterior $p_\theta(z|x)$ to generate the latent variable given a set of data x. to obtain this, we need

$$p_\theta(z|x) = \frac{p_\theta(x|z)p(z)}{p_\theta(x)}$$

but this return us back to the problem that we don't have the direct access to $p(x)$ and this term is intractable. To get around this, we introduce the encoder $q_\phi(z|x)$ that aims to parameterize the true posterior $p(z|x)$.

Variational Autoencoders (4)

Now, to train the parameterized models, we aim to maximize the log likelihood of the distribution:

$$\log p(\mathbf{x})$$

In the following we show how this term is equivalent to

$$\log p(\mathbf{x}) = \mathbb{E}_{q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})}\left[\log \frac{p(\mathbf{x},\mathbf{z})}{q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})}\right] + D_{\mathrm{KL}}(q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})||p(\mathbf{z}|\mathbf{x}))$$

Variational Autoencoders (5)

$$\log p(\mathbf{x}) = \log p(\mathbf{x}) \int q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x}) d\mathbf{z} \tag{1}$$

$$= \int \log p(\mathbf{x}) q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x}) d\mathbf{z} \tag{2}$$

$$= \mathbb{E}_{q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})} \left[\log p(\mathbf{x}) \right] \tag{3}$$

$$= \mathbb{E}_{q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})} \left[\log \frac{p(\mathbf{x}, \mathbf{z})}{p(\mathbf{z}|\mathbf{x})} \right] \tag{4}$$

$$= \mathbb{E}_{q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})} \left[\log \frac{p(\mathbf{x}, \mathbf{z})}{p(\mathbf{z}|\mathbf{x})} \frac{q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})}{q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})} \right] \tag{5}$$

$$= \mathbb{E}_{q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})} \left[\log \frac{p(\mathbf{x}, \mathbf{z})}{q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})} \right] + \mathbb{E}_{q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})} \left[\log \frac{q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})}{p(\mathbf{z}|\mathbf{x})} \right] \tag{6}$$

$$= \mathbb{E}_{q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})} \left[\log \frac{p(\mathbf{x}, \mathbf{z})}{q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})} \right] + D_{\mathrm{KL}}(q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x}) || p(\mathbf{z}|\mathbf{x})) \tag{7}$$

Variational Autoencoders (6)

Since D_{KL} is a non-negative metric, we now have the term as the lower bound of the evidance (ELBO)

$$\mathrm{ELBO} = \mathbb{E}_{q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})}\left[\log \frac{p(\mathbf{x}, \mathbf{z})}{q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})}\right]$$

By maximizing ELBO, we are minimizing the KL divergence between the parameterized encoder $q_{\phi}(\mathbf{z}|\mathbf{x})$ and the true conditional distribution $p(\mathbf{z}|\mathbf{x})$.

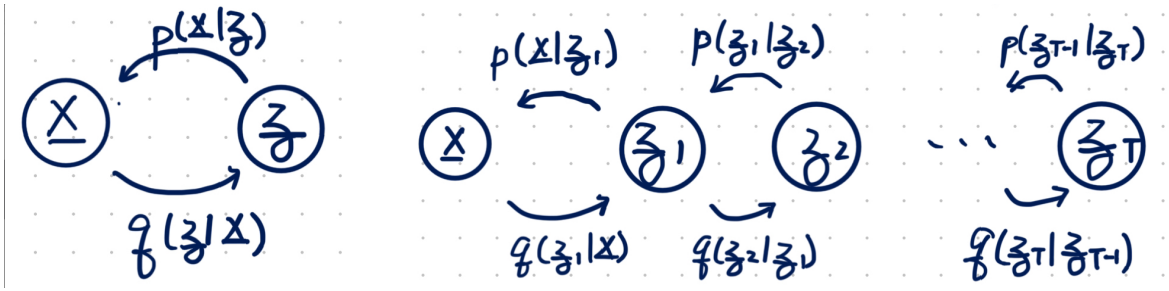
Variational Autoencoders (7)

If we further expand the ELBO

$$\begin{aligned}
\text{ELBO} &= \mathbb{E}_{q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})}\left[\log \frac{p_\theta(\mathbf{x}|\mathbf{z})p(\mathbf{z})}{q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})}\right] \\
&= \mathbb{E}_{q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})}\left[\log \frac{p_\theta(\mathbf{x}|\mathbf{z})p(\mathbf{z})}{q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})}\right] \\
&= \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}\big[\log p_\theta(\mathbf{x} \mid \mathbf{z})\big] \;-\; D_{\mathrm{KL}}\big(q_\phi(\mathbf{z} \mid \mathbf{x}) \,\|\, p(\mathbf{z})\big).
\end{aligned}$$

Hierarchical Variational Autoencoders (1)

the latent variable of VAE is extended to multiple hierarchies T to form a HVAE, where each level of the latent variables are modeled by a higher level latents.



(a) VAE (b) HVAE

Hierarchical Variational Autoencoders (2)

We usually assume the transition along hierarchy is Markovian, then the joint distribution of data $\mathbf{x}$ and all latents $\mathbf{z}_{1:T}$ can therefore be written as

$$p_\theta(\mathbf{x}, \mathbf{z}_{1:T}) = p(\mathbf{z}_T) p_\theta(\mathbf{x}|\mathbf{z}_1) \prod_{t=2}^{T} p_\theta(\mathbf{z}_{t-1}|\mathbf{z}_t),$$

and the posterior is

$$q_{\boldsymbol{\phi}}(\mathbf{z}_{1:T}|\mathbf{x}) = q_{\boldsymbol{\phi}}(\mathbf{z}_1|\mathbf{x}) \prod_{t=2}^{T} q_{\boldsymbol{\phi}}(\mathbf{z}_t|\mathbf{z}_{t-1})$$

Hierarchical Variational Autoencoders (3)

The ELBO just add the KL divergence terms for the latent trajectories in HVAE,

$$\begin{aligned}\log p_\theta(\mathbf{x}_0) \;&\geq\; \text{ELBO} \\ &= \mathbb{E}_q\Big[\log p_\theta(\mathbf{x}_0 \mid \mathbf{z}_1)\Big] - \sum_{t=2}^{T} D_{\mathrm{KL}}\big(q_\phi(\mathbf{z}_{t-1} \mid \mathbf{z}_t, \mathbf{x}_0) \,\|\, p_\theta(\mathbf{z}_{t-1} \mid \mathbf{z}_t)\big) \\ &\quad - D_{\mathrm{KL}}\big(q_\phi(\mathbf{z}_T \mid \mathbf{x}_0) \,\|\, p(\mathbf{z}_T)\big).\end{aligned}$$

we include the ELBO of the VAE here for an easy comparison.

$$\mathbb{E}_{q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})}\left[\log \frac{p_{\boldsymbol{\theta}}(\mathbf{x},\mathbf{z})}{q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})}\right] = \mathbb{E}_{q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})}\left[\log p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z})\right] - D_{\mathrm{KL}}(q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})||p_{\boldsymbol{\theta}}(\mathbf{z})) \tag{8}$$

Diffusion models fix the forward process to a conditional Gaussian,

$$q(x_t \mid x_{t-1}) = \mathcal{N}\left(\sqrt{1-\beta_t}\, x_{t-1},\ \beta_t I\right)$$

which is exactly the infinitesimal noise we described at the beginning.

We directly adapt the objective from HVAE,

$$\mathcal{L}_{\text{DDPM}} = \mathbb{E}_q[\log p_\theta(\mathbf{x}_0 \mid \mathbf{x}_1)] - \sum_{t=2}^{T} D_{\text{KL}}(q(\mathbf{x}_{t-1} \mid \mathbf{x}_t, \mathbf{x}_0) \,\|\, p_\theta(\mathbf{x}_{t-1} \mid \mathbf{x}_t)) - D_{\text{KL}}(q(\mathbf{x}_T \mid \mathbf{x}_0) \,\|\, p(\mathbf{x}_T))$$

Objective with Gaussian Assumption (1)

Since our goal is to let the model predicting the noise distribution, we design $p_{\boldsymbol{\theta}}$ as a Gaussian which is naturally parameterized by the two variables,

- for $\boldsymbol{\Sigma}_p$, since variance in forward process q is given, we set $\boldsymbol{\Sigma}_p(t) = \boldsymbol{\Sigma}_q(t) = \sigma_q^2(t)\mathbf{I}$
- we parameterize the mean with $\boldsymbol{\mu}_p = \boldsymbol{\mu}_{\boldsymbol{\theta}}(\mathbf{x}_t, t)$

since now the forward q and the reverse $p_{\boldsymbol{\theta}}$ are assumed to be Gaussian, the KL divergence

$$D_{\mathrm{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)||p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t))$$

can be simplified with the equation

$$D_{\mathrm{KL}}(\mathcal{N}(\mathbf{x}_1; \boldsymbol{\mu}_{x_1}, \boldsymbol{\Sigma}_{x_1})||\mathcal{N}(\mathbf{x}_2; \boldsymbol{\mu}_{x_2}, \boldsymbol{\Sigma}_{x_2}))$$

$$= \frac{1}{2}\left[\log\frac{|\boldsymbol{\Sigma}_{x_2}|}{|\boldsymbol{\Sigma}_{x_1}|} - \dim(\mathbf{x}) + \mathrm{tr}(\boldsymbol{\Sigma}_{x_2}^{-1}\boldsymbol{\Sigma}_{x_1}) + (\boldsymbol{\mu}_{\mathbf{x}_2} - \boldsymbol{\mu}_{\mathbf{x}_1})^T\boldsymbol{\Sigma}_{x_2}^{-1}(\boldsymbol{\mu}_{\mathbf{x}_2} - \boldsymbol{\mu}_{\mathbf{x}_1})\right]$$

Objective with Gaussian Assumption (2)

With again, some tedious arithmetics, minimizing the KL divergence can be shown to be equal to minimizing the MSE of the estimated mean.

$$\begin{aligned} &\arg\min_{\boldsymbol{\theta}} \; D_{\mathrm{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)||p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t)) \\ = &\arg\min_{\boldsymbol{\theta}} \; \frac{1}{2\sigma_q^2(t)} \left[||(\boldsymbol{\mu}_{\boldsymbol{\theta}} - \boldsymbol{\mu}_q)||^2\right] \end{aligned}$$

Objective with Gaussian Assumption (3)

With some further tedious arithmetics one can show that minimizing the objective is equivalent to minimize the following two objectives

$$\arg\min_{\boldsymbol{\theta}} \frac{1}{2\sigma_q^2(t)} \frac{\sqrt{\bar{\alpha}_{t-1}}(1-\alpha_t)}{1-\bar{\alpha}_{t-1}} \left[||\hat{\mathbf{x}}_{\boldsymbol{\theta}}(\mathbf{x}_t, t) - \mathbf{x}_0||_2^2\right]$$

$$\arg\min_{\boldsymbol{\theta}} \frac{1}{2\sigma_q^2(t)} \frac{(1-\alpha_t)^2}{(1-\alpha_t)\alpha_t} [\|\boldsymbol{\epsilon}_0 - \hat{\boldsymbol{\epsilon}}_\theta(\mathbf{x}_t, t)\|_2^2]$$

Through the following two equalities.

$$\boldsymbol{\mu}_q(\mathbf{x}_t, \mathbf{x}_0) = \frac{1}{\sqrt{\alpha_t}}\mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}\sqrt{\alpha_t}}\boldsymbol{\epsilon}_0$$

$$\mathbf{x}_0 = \frac{\mathbf{x}_t - \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}_0}{\sqrt{\bar{\alpha}_t}}$$

Describing Evolution

Several familiar ways to describe such an Evolution

- **Random Process**. Directly describing the states $\mathbf{x}(t) = \mathbf{X}_t$, where at each fixed $t \in [0, 1]$ the mapping is a random variable.
- **Density evolution**. Similarly, specify each $\{p_t\}$ such that $\mathbf{X}_t \sim p_t$.
- **ODE**. Find a velocity field acting on the states,

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{v}(\mathbf{x}_t, t)$$

  so that for each time instances $\mathbf{x}_t \sim p_t$
- **Stochastic Differential Equations (SDE)**.

For an EE-friendly environment, we try to bridge us from ODE to SDE in the next few slides.

Deterministic to Stochastic Differential Equations (1)

The following tries to give an intuitive transition from ODE to SDE.

Let us start from an ODE. Consider a deterministic evolution of a state variable $\mathbf{x}(t) \in \mathbb{R}^D$

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{f}(\mathbf{x}_t, t)$$

If we observe the dynamics over a small time step Δt we can express it in a discretized mapping,

$$\mathbf{x}_{t+\Delta t} = \mathbf{x}_t + \mathbf{f}(\mathbf{x}_t, t)\Delta t$$

What if there exists a noise term?

$$\mathbf{x}_{t+\Delta t} = \mathbf{x}_t + \mathbf{f}(\mathbf{x}_t, t)\Delta t + g(t)\sqrt{\Delta t}\boldsymbol{\epsilon}_t$$

Taking the limit this give us a stochastic differential equation

$$d\mathbf{x}(t) = f(\mathbf{x}(t), t)\, dt + g(t)\, d\mathbf{w}(t).$$

SDE and its induced Density Evolution

Fokker-Planck Equation

Suppose the dynamics of X_t can be described by a forward SDE

$$dX_t = f(X_t, t)\, dt + g(t)\, dW_t.$$

then the Fokker–Planck equation gives the PDE of the marginal

$$\partial_t p_t(x) = -\nabla \cdot \big(f(x,t)\, p_t(x)\big) + \tfrac{1}{2} g^2(t)\, \Delta p_t(x).$$

This is essentially the SDE version *change of variables*.

The following sections try to give a walk through the proof.

Change of Variables

If we recall our *Probability*, we're already familiar with the density distribution after a diterministic mapping of a random variable.

Derived Distribution (Discrete Transition)

Suppose a random variable $\mathbf{x}_0 \sim p_0$ was mapped by a deterministic state transition function $\boldsymbol{\Psi} : \mathbb{R}^n \to \mathbb{R}^n$.

$$\mathbf{x}_1 = \boldsymbol{\Psi}(\mathbf{x}_0).$$

Then the pdf of $\mathbf{x}_1$ is

$$p_1(\mathbf{x}_1) = p_0(\boldsymbol{\Psi}^{-1}(\mathbf{x}_1)) \left| \det \left(\frac{\partial \boldsymbol{\Psi}^{-1}}{\partial \mathbf{x}_1} \right) \right| = p_0(\boldsymbol{\Psi}^{-1}(\mathbf{x}_1)) \left| \det \left(\frac{\partial \boldsymbol{\Psi}}{\partial \mathbf{x}_0} \right) \right|^{-1}.$$

Continuous State Transition

Let us now consider a continuous state transition described by an ODE,

$$d\mathbf{x}_t = \mathbf{f}(\mathbf{x}_t, t)\, dt.$$

where $\mathbf{f} : \mathbb{R}^n \times [0, T] \to \mathbb{R}^n$.

For a state transition following the above ODE, the probability distribution of $\mathbf{x}$ evolves as

$$\frac{\partial}{\partial t} p_t(\mathbf{x}) = -\nabla \cdot (p_t(\mathbf{x}) \mathbf{f}(\mathbf{x}, t)).$$

The proof rewrites the ODE into infinitesimal deterministic mappings, find the Jacobian, and then take limits. We leave the proof for the mathematically-inclined audience in the appendix 55.

Density Evolution of a Continuous State Transition (1)

Continuity Equation

Let $\mathbf{f} : \mathbb{R}^n \times [0, T] \to \mathbb{R}^n$ be a time-dependent mapping satisfying

- for each time slices $f(\cdot, t) \in C^1$
- f is locally Lipschitz in x

Consider the states $\mathbf{x}_t$,$t \in [0, T]$ defined by the deterministic state transition,

$$d\mathbf{x}_t = \mathbf{f}(\mathbf{x}_t, t)\, dt,$$

with a given initial state $\mathbf{x}_0 \sim p_0$.

Assume that for each time slice t, the law of the $\mathbf{x}_t$ follows a nice density p_t. Then the family of the densities of all t satisfies the Continuity Equation

$$\frac{\partial}{\partial t} p_t(\mathbf{x}) = -\nabla_{\mathbf{x}} \cdot (p_t(\mathbf{x}) \mathbf{f}(\mathbf{x}, t)).$$

Density Evolution of a Continuous State Transition (2)

Let us try to inspect its physical intuition.

- the overall "probability mass" is conservative.
- the drift function (the velocity field) acts on the particle $\mathbf{x}_t$ only by re-distributing them in space.
- the divergence $\nabla_{\mathbf{x}} \cdot (p_t(\mathbf{x})\mathbf{f}(\mathbf{x}, t))$ described how the flow stretches or compresses the density
- correspondingly the density changes to preserve the total probability.

Denoising with ODE (1)

At the very beginning where we construct the forward diffusion by adding the conditional noise

$$p_t(\mathbf{x}_t|\,\mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \alpha_t \mathbf{x}_0, \sigma_t^2 \mathbf{I})$$

Since all the nice Gaussianity, while conditioning on $\mathbf{x}_0$ the corresponding random process is

$$\mathbf{x}_t = \alpha_t \mathbf{x}_0 + \sigma_t \boldsymbol{\epsilon},$$

The corresponding ODE description of it would therefore be

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{v}_t(\mathbf{x}(t)), \quad \mathbf{v}_t(\mathbf{x}) = \mathbb{E}[\alpha_t' + \sigma_t' | \mathbf{x}]$$

Denoising with ODE (2)

So the variational-based "data reconstructing" or "noise predicting" model designs are actually equivelant to learning the velocity field. This was called *Flow Matching* and is first proposed by

$$\mathcal{L}_{\text{FM}}(\phi) \;=\; \mathbb{E}_{t,\,\mathbf{x}_t\sim p_t}\left[\left\|\mathbf{v}_\phi(\mathbf{x}_t,t)-\mathbf{v}_t(\mathbf{x}_t)\right\|_2^2\right]$$

Although the true velocity is not directly accessible, one can find out that the conditional flow matching objective is equivalent to flow matching with a constant difference,

$$\begin{aligned}\mathcal{L}_{\text{CFM}}(\phi) \;&=\; \mathbb{E}_{t,\,\mathbf{x}_0,\mathbf{x}_1\sim\pi(\mathbf{x}_0,\mathbf{x}_1),\,\mathbf{x}\sim p_t(\cdot|\mathbf{x}_0,\mathbf{x}_1)}\left[\left\|\mathbf{v}_\phi(\mathbf{x}_t,t)-\mathbf{v}_t(\mathbf{x}_t\mid\mathbf{x}_0,\mathbf{x}_1)\right\|_2^2\right]\\ &= \mathcal{L}_{\text{FM}}(\phi)+\text{Const.}\end{aligned}$$

Deterministic vs. Stochastic

We try to bridge ourselves from deterministic to stochastic version of the derived distribution.

	Deterministic	**Stochastic**
$d\mathbf{x}_t =$	$\mathbf{f}(\mathbf{x}_t, t)dt$	$\mathbf{f}(\mathbf{x}_t, t)\,dt + g(t)\,dW_t$
$\partial_t p_t =$	$-\nabla_{\mathbf{x}} \cdot \big(p_t(\mathbf{x})\mathbf{f}(\mathbf{x}, t)\big)$	$-\nabla_{\mathbf{x}} \cdot \big[p_t(\mathbf{x})\big(\mathbf{f}(\mathbf{x}, t) + \frac{1}{2}g^2(t)\nabla_{\mathbf{x}} \log p_t(\mathbf{x})\big)\big]$

The derived distribution of a stochastic evolution has an extra second order diffusion term.

Notice that the second-order of the diffusion term

$$(g(t)\,dW_t)^2$$

is in fact of order $O(dt)$ because $d\mathbf{w}_t \sim \mathcal{N}(0, \sqrt{dt}\mathbf{I})$.

Density Evolution of an SDE

Density Evolution of an SDE (the Fokker–Planck Equation)

Consider the evolution of states $\mathbf{x}_t, t \in [0, T]$ defined by an SDE,

$$d\mathbf{x}_t = \mathbf{f}(\mathbf{x}_t, t)\, dt + g(t) d\mathbf{w}_t,$$

with a given initial state $\mathbf{x}_0 \sim p_0$.

Assume that for each time slice t, the law of the $\mathbf{x}_t$ follows a nice density p_t. Then the family of the densities of all t satisfies the Fokker–Planck Equation,

$$\frac{\partial}{\partial t} p_t(\mathbf{x}) = -\nabla_{\mathbf{x}} \cdot \left[p_t(\mathbf{x}) \left(\mathbf{f}(\mathbf{x}, t) + \frac{1}{2} g^2(t) \nabla_{\mathbf{x}} \log p_t(\mathbf{x}) \right) \right].$$

We leave the proof in the appendix 55 for those who are interested in it.

from Foward to Reverse SDE (1)

For ODE, we can directly integrate it reversely to obtain $\mathbf{x}_0$. How to achieve that with an SDE?

Reverse-Time SDE (Anderson, 1982)

Let $\{\mathbf{x}(t)\}_{t\in[0,T]}$ follow the forward-time SDE

$$d\mathbf{x}(t) = \mathbf{f}(\mathbf{x}(t), t)\, dt + g(t)\, d\mathbf{w}(t),$$

with marginal density $p_t(x)$. Then the time-reversed process $\{\bar{\mathbf{x}}(t)\}_{t\in[0,T]}$, evolving from time T to 0, satisfies the reverse-time SDE

$$d\bar{\mathbf{x}}(t) = \Big[\mathbf{f}(\bar{\mathbf{x}}(t), t) - g^2(t)\, \nabla_{\mathbf{x}} \log p_t(\bar{\mathbf{x}}(t))\Big]\, dt + g(t)\, d\bar{\mathbf{w}}(t),$$

where $\bar{\mathbf{w}}(t)$ is a standard Wiener process evolving in reverse time.

from Foward to Reverse SDE (2)

The proof starts from the Forward SDE

$$d\mathbf{x}(t) = \mathbf{f}(\mathbf{x}, t)\, dt + g(t)\, d\mathbf{w}(t).$$

We've already shown that the Fokker–Planck equation gives the PDE for the marginal:

$$\partial_t q_t(\mathbf{x}) = -\nabla \cdot \big(\mathbf{f}(\mathbf{x}, t)\, q_t(\mathbf{x})\big) + \tfrac{1}{2} g^2(t)\, \Delta q_t(\mathbf{x}).$$

from Foward to Reverse SDE (3)

Reverse process: Let $\bar{\mathbf{x}}_\tau = \mathbf{x}_{T-\tau}$ with marginal $q_{T-\tau}(\mathbf{x})$. Then

$$\partial_\tau q_{T-\tau}(\mathbf{x}) = -\partial_t q_t(\mathbf{x})\big|_{t=T-\tau}.$$

Fokker–Planck of the reversed process becomes

$$\partial_\tau q_{T-\tau}(\mathbf{x}) = \nabla \cdot \big(\mathbf{f}(\mathbf{x}, t)\, q_{T-\tau}(\mathbf{x})\big) - \tfrac{1}{2} g^2(t)\, \Delta q_{T-\tau}(\mathbf{x}).$$

Suppose a reverse SDE exists with this given marginal,

$$d\bar{\mathbf{x}}_t = \mathbf{f}_{\text{rev}}(\bar{\mathbf{x}}_t, t)\, dt + g(t)\, d\bar{\mathbf{w}}_t,$$

according to Fokker-Planck again, the $q_{T-\tau}(x)$ satisfies

$$\partial_\tau q_{T-\tau}(\mathbf{x}) = -\nabla \cdot \big(\mathbf{f}_{\text{rev}}(\mathbf{x}, t)\, q_{T-\tau}(\mathbf{x})\big) + \tfrac{1}{2} g^2(t)\, \Delta q_{T-\tau}(\mathbf{x}).$$

Thus, solving for $\mathbf{f}_{\text{rev}}(\mathbf{x}, t)$ gives the drift of the reverse process.

Reverse SDE (Anderson, 1982) (1)

From equating the Fokker–Planck equations, we obtain

$$\nabla \cdot \big(\mathbf{f}(\mathbf{x}, t) q_t(x)\big) - \tfrac{1}{2} g^2(t) \Delta q_t(\mathbf{x}) = -\nabla \cdot \big(\mathbf{f}_{\text{rev}}(\mathbf{x}, t) q_t(\mathbf{x})\big) + \tfrac{1}{2} g^2(t) \Delta q_t(\mathbf{x}).$$

Therefore,

$$\nabla \cdot \big(\mathbf{f}_{\text{rev}}(\mathbf{x}, t) q_t(\mathbf{x})\big) = \nabla \cdot \big(\mathbf{f}(\mathbf{x}, t) q_t(\mathbf{x})\big) - g^2(t)\, \Delta q_t(\mathbf{x}).$$

using the identity

$$\Delta q_t(\mathbf{x}) = \nabla \cdot \nabla q_t(\mathbf{x}) = \nabla \cdot \big(q_t(\mathbf{x}) \nabla \log q_t(\mathbf{x})\big).$$

We can see that if we let the reverse drift function to be

$$\mathbf{f}_{\text{rev}}(\mathbf{x}, t) = \mathbf{f}(\mathbf{x}, t) - g^2(t)\, \nabla_{\mathbf{x}} \log q_t(\mathbf{x}).$$

then the divergence would be matching.

Reverse SDE (Anderson, 1982) (2)

We conclude our **Reverse-time SDE** to be

$$d\mathbf{x}_t = \Big(\mathbf{f}(\mathbf{x}_t, t) - g^2(t)\,\nabla_{\mathbf{x}} \log q_t(\mathbf{x}_t)\Big)\,dt + g(t)\,d\bar{\mathbf{w}}_t.$$

- time is now running reversely as $t = T - \tau$
- Drift contains the original $\mathbf{f}(\mathbf{x}, t)$ plus a correction term.
- The correction involves the **score** function $\mathbf{s}(\mathbf{x}, t) = \nabla_{\mathbf{x}} \log q_t(\mathbf{x})$.
- This is the mathematical foundation for score-based generative models.

Turning Back the Time

Reversing Density Evolution

Suppose the dynamics of $\mathbf{x}_t$ can be described by a forward SDE

$$d\mathbf{x}_t = \mathbf{f}(\mathbf{x}_t, t)\, dt + g(t)\, d\mathbf{w}_t.$$

Then the following two equations yield the same marginals $\{p_t\}_{t\in[0,1]}$

- **Probability-Flow ODE**.

$$\frac{d\tilde{\mathbf{x}}(t)}{dt} = \mathbf{v}\big(\tilde{\mathbf{x}}(t), t\big), \quad \mathbf{v}(\mathbf{x}, t) = \mathbf{f}(\mathbf{x}, t) - \frac{1}{2}\, g^2(t)\, \nabla_x \log p_t(x).$$

- **Reverse-Time SDE**.

$$d\bar{\mathbf{x}}(t) = \Big[\mathbf{f}(\bar{\mathbf{x}}(t), t) - g^2(t)\, \nabla_{\mathbf{x}} \log p_t(\bar{\mathbf{x}}(t))\Big]\, dt + g(t)\, d\bar{\mathbf{w}}(t).$$

Denoising with SDE (1)

We know from our previous derivations that the velocity and the score are related as

$$\mathbf{v}(\mathbf{x}, t) \;=\; \mathbf{f}(\mathbf{x}, t) \;-\; \frac{1}{2}\, g^2(t)\, \mathbf{s}(\mathbf{x}, t).$$

We can show that the reverse-time SDEs that generates the same marginals are

$$d\bar{\mathbf{x}}(t) = [\mathbf{v}(\bar{\mathbf{x}}(t), t) - \frac{1}{2}\gamma^2 \mathbf{s}(\bar{\mathbf{x}}(t), t)]\, d\bar{t} + \gamma(t)\, d\bar{\mathbf{w}}(t)$$

for arbitrary non-negative coefficient $\gamma(t)$. We easily verify the SDE density evolution by checking the Fokker-Planck equation of it,

$$\partial_t p = -\nabla_{\mathbf{x}} \cdot \left([\mathbf{v} - \frac{1}{2}\gamma^2 \mathbf{s}]p\right) + \frac{1}{2}\gamma \Delta p$$

Denoising with SDE (2)

and realize that it is equivalent to the continuity equation of the given ODE,

$$
\begin{aligned}
&-\nabla_{\mathbf{x}}\cdot\big([\mathbf{v}-\frac{1}{2}\gamma^2\mathbf{s}]p\big)+\frac{1}{2}\gamma\Delta p \\
&\quad=-\nabla_{\mathbf{x}}\cdot(\mathbf{v}p)-\frac{1}{2}\gamma^2\nabla_{\mathbf{x}}\cdot(\mathbf{s}p)+\frac{1}{2}\gamma\Delta p \\
&\quad=-\nabla_{\mathbf{x}}\cdot(\mathbf{v}p)-\frac{1}{2}\gamma^2\nabla_{\mathbf{x}}\cdot(p\nabla\log p)+\frac{1}{2}\gamma\Delta p \\
&\quad=-\nabla_{\mathbf{x}}\cdot(\mathbf{v}p)-\frac{1}{2}\gamma^2\nabla_{\mathbf{x}}\cdot(\nabla p)+\frac{1}{2}\gamma\Delta p \\
&\quad=-\nabla_{\mathbf{x}}\cdot(\mathbf{v}p)-\frac{1}{2}\gamma^2\Delta p+\frac{1}{2}\gamma\Delta p \\
&\quad=-\nabla_{\mathbf{x}}\cdot(\mathbf{v}p)
\end{aligned}
$$

DDPM as a discretized SDE (1)

We want to express a discrete Markov chain (DDPM) as a discretized forward SDE.

$$dX_t = f(X_t, t)\,dt + g(t)\,dW_t.$$

Starts from our DDPM of one timestep,

$$x_i = \sqrt{1-\beta_i}\,x_{i-1} + \sqrt{\beta_i}\,\epsilon_{i-1}, \qquad \epsilon_{i-1} \sim \mathcal{N}(0, I).$$

Let $\tilde{\beta}_i = N \cdot \beta_i$ and rewrite:

$$x_i = \sqrt{1-\frac{\tilde{\beta}_i}{N}}\,x_{i-1} + \sqrt{\frac{\tilde{\beta}_i}{N}}\,\epsilon_{i-1}.$$

DDPM as a discretized SDE (2)

In continuous time we consider

$$\tilde{\beta}_i \to \tilde{\beta}(t = \frac{i}{N})$$
$$\epsilon_i \to z(t = \frac{i}{N})$$
$$x_t \to X(t = \frac{i}{N})$$

as $N \to \infty$, and by letting $t = \{0, \frac{1}{N}, \ldots, \frac{N-1}{N}\}$, $\Delta t = \frac{1}{N}$

$$X(t + \Delta t) = \sqrt{1 - \tilde{\beta}(t)\Delta t}\, X(t) + \sqrt{\tilde{\beta}(t)\Delta t}\, z(t), \quad z(t) \sim \mathcal{N}(0, I).$$

DDPM as a discretized SDE (3)

using the approximation with small x,

$$\sqrt{1-x} \approx 1 - \tfrac{1}{2}x.$$

and for small δt we have $\tilde{\beta}(t+\delta t) = \tilde{\beta}(t)$, therefore we have,

$$X(t+\Delta t) - X(t) \;\approx\; -\tfrac{1}{2}\tilde{\beta}(t)\, X(t)\, \Delta t + \sqrt{\tilde{\beta}(t)}\, z(t)\sqrt{\Delta t}.$$

Taking the limit,

$$dX_t = -\tfrac{1}{2}\beta(t)\, X_t\, dt + \sqrt{\beta(t)}\, dW_t.$$

This is exactly an SDE with affine drift $f(X_t, t) = -\frac{1}{2}\beta(t)\, X_t$ and diffusion $g(t) = \sqrt{\beta(t)}$ coefficient.

From noise MSE to score matching

Starting from our noise MSE objective of DDPM,

$$\arg\min_{\theta} D_{\mathrm{KL}}(q(x_{t-1} \mid x_t, x_0) \,\|\, p_\theta(x_{t-1} \mid x_t))$$

$$= \arg\min_{\theta} \frac{1}{2\sigma_q^2(t)} \frac{(1-\alpha_t)^2}{(1-\bar{\alpha}_t)\alpha_t} \|\epsilon_0 - \hat{\epsilon}_\theta(x_t, t)\|^2.$$

Score of $q(x_t \mid x_0)$ is the gradient of the log likelihood

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1-\bar{\alpha}_t}\,\epsilon_0, \quad q(x_t \mid x_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t} x_0, (1-\bar{\alpha}_t) I).$$

$$\nabla_{x_t} \log q(x_t \mid x_0) = -\frac{1}{\sqrt{1-\bar{\alpha}_t}} \epsilon_0.$$

therefore the model predicting the noise can also be seen as a score predictor

$$s_\theta(x_t, t) := -\frac{1}{\sqrt{1-\bar{\alpha}_t}}\, \hat{\epsilon}_\theta(x_t, t).$$

$$\|\epsilon_0 - \hat{\epsilon}_\theta(x_t, t)\|^2 = (1-\bar{\alpha}_t) \|\nabla_{x_t} \log q(x_t \mid x_0) - s_\theta(x_t, t)\|^2.$$

Thus DDPM noise MSE $\Leftrightarrow$ *denoising score matching*.

Four Parameterizations in Diffusion Models (1)

Throughout the discussion we see that for a forward diffusion process,

$$p_t(x_t \mid x_0) \;=\; \mathcal{N}\big(x_t;\, \alpha_t x_0,\, \sigma_t^2 I\big).$$

with the initial condition as the data distribution $x_0 \sim p_{\text{data}}$, we have several different ways to model this dynamics.

Four Parameterizations in Diffusion Models (2)

Variational. Starting from the Divergence analysis we arrived at,

i. ϵ-Prediction (Noise Prediction).

$$\epsilon_\phi(x_t, t) \approx \mathbb{E}[\epsilon \mid x_t], \quad \mathcal{L}_{\text{noise}}(\phi) := \mathbf{E}_t\Big[\omega(t)\, \mathbf{E}_{x_0,\epsilon}\big\|\epsilon_\phi(x_t, t) - \epsilon\big\|_2^2\Big].$$

ii. x-Prediction (Clean Prediction).

$$x_\phi(x_t, t) \approx \mathbb{E}[x_0 \mid x_t], \quad \mathcal{L}_{\text{clean}}(\phi) := \mathbf{E}_t\Big[\omega(t)\, \mathbf{E}_{x_0,\epsilon}\big\|x_\phi(x_t, t) - x_0\big\|_2^2\Big].$$

Four Parameterizations in Diffusion Models (3)

3. Score Prediction.

This approach predicts the score function at noise level t, which points in the average direction that denoises x_t toward all possible clean samples that could have generated it.

The estimator is defined as

$$s_\phi(x_t, t) \approx \nabla_{x_t} \log p_t(x_t) = \mathbb{E}[\nabla_{x_t} \log p_t(x_t \mid x_0) \mid x_t] = s^\star(x_t, t).$$

The training objective is

$$\mathcal{L}_{\text{score}}(\phi) := \mathbb{E}_t \Big[\omega(t) \, \mathbb{E}_{\mathbf{x}_0, \epsilon} \big\| \mathbf{s}_\phi(\mathbf{x}_t, t) - \nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t \mid \mathbf{x}_0) \big\|_2^2 \Big].$$

The conditional score satisfies

$$\nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t \mid \mathbf{x}_0) = -\frac{1}{\sigma_t} \epsilon.$$

Four Parameterizations in Diffusion Models (4)

4. v-Prediction (Flow-based). This perspective predicts the instantaneous average velocity of the data as it evolves through $\mathbf{x}_t$. The estimator is defined as

$$\mathbf{v}_\phi(\mathbf{x}_t, t) \;\approx\; \mathbb{E}\left[\frac{d\mathbf{x}_t}{dt} \;\middle|\; \mathbf{x}_t\right] \;=\; \mathbf{v}^\star(\mathbf{x}_t, t).$$

The training objective is

$$\mathcal{L}_{\text{velocity}}(\phi) \;:=\; \mathbb{E}_t\Big[\omega(t)\,\mathbb{E}_{\mathbf{x}_0,\epsilon}\big\|\mathbf{v}_\phi(\mathbf{x}_t, t) - \mathbf{v}_t(\mathbf{x}_t \mid x_0, \epsilon)\big\|_2^2\Big].$$

The conditional velocity is given by

$$\mathbf{v}_t(\mathbf{x}_t \mid \mathbf{x}_0, \epsilon) \;=\; \alpha'_t\mathbf{x}_0 + \sigma'_t\epsilon.$$

Unifying Perspective

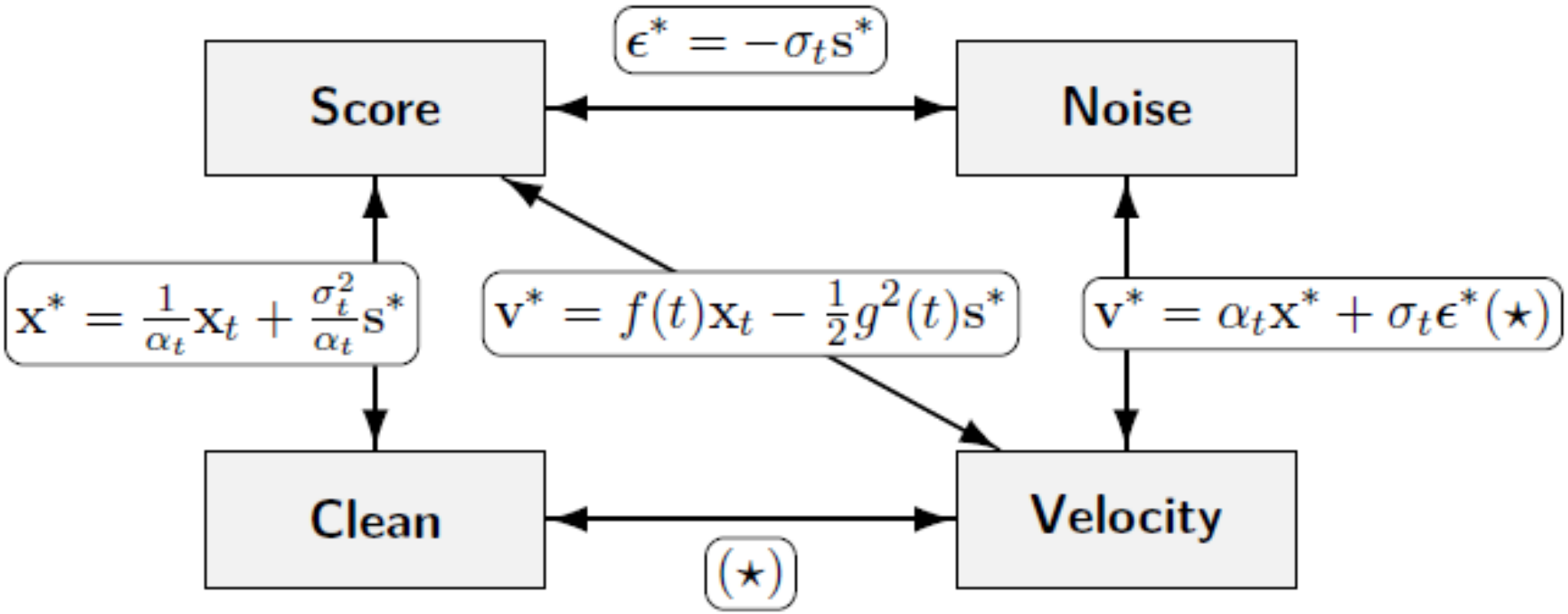


Figure: An unified perspective of the four different parameterizations.

Engineering Perspectives

Controlling the Path

- **Score-based (SDE)**: Conditioning as guidance or force fields. Soft guidance.
- **Flow-based (ODE)**: Conditioning via deterministic control and geometry. Hard constraints.
- **Variational**: Conditioning through latent variables

Numerical

- **Score-based**: Cheap training, expensive sampling
- **Flow-based**: Structured training, deterministic inference
- **Variational**: Fast inference, no iterative solvers

Failure Modes

- **Score-based**: Instability near low-noise regimes
- **Flow-based**: Geometric errors cause trajectory collapse
- **Variational**: Posterior collapse and over-smoothing

Choosing Your Favorite

Variational

- One-shot and fast inference
- Latent representations for downstream tasks
- Stable quality

Score-Based

- Highest sample quality and diversity
- Flexible guidance and conditioning
- Long or stochastic sampling

Flow / Velocity-Based

- Deterministic and coherent mappings are desired
- Geometric structure or transport interpretation
- Predictable inference time

Modeling Philosophy Behind Diffusion Parameterizations

- **Variational Diffusion**: Learn a latent representation for fast, amortized inference
- **Score-Based Diffusion**: Learn the local geometry of the data distribution via gradients
- **Flow-Based Diffusion**: Learn a global transport or dynamical system connecting distributions

Derived Distribution of an ODE (1)

The proof starts from equivalently interpret the state transition ODE as small bijective updates

$$\mathbf{x}_{t+\Delta t} = \boldsymbol{\Psi}(\mathbf{x}_t) \coloneqq \mathbf{x}_t + \Delta t \mathbf{f}(\mathbf{x}_t, t). \tag{9}$$

Treating it as nice deterministic function as in the previous section, we find its Jacobian,

$$\begin{aligned}
\frac{\partial \Psi}{\partial \mathbf{x}_t} &= I + \Delta t \, \nabla_{\mathbf{x}} f(\mathbf{x}_t, t) + O(\Delta t^2), \\
\det\left(\frac{\partial \Psi}{\partial \mathbf{x}_t}\right) &= \det(I + \Delta t \, \nabla_{\mathbf{x}} f(\mathbf{x}_t, t)) + O(\Delta t^2) \\
&\simeq \left(1 + \Delta t \, \mathrm{Tr}\left(\nabla_{\mathbf{x}} f(\mathbf{x}_t, t)\right) + O(\Delta t^2)\right) + O(\Delta t^2) \\
&= 1 + \Delta t \, \nabla_{\mathbf{x}} \cdot f(\mathbf{x}_t, t) + O(\Delta t^2).
\end{aligned}$$

trace of gradient is the divergence.

Derived Distribution of an ODE (2)

And thus the log density follows

$$\begin{aligned}\log p_{t+\Delta t}(x_{t+\Delta t}) &= \log p_t(\mathbf{x}_t) - \log(1 + \Delta t\, \nabla_{\mathbf{x}} \cdot f(\mathbf{x}_t, t)) + O(\Delta t^2) \\ &= \log p_t(\mathbf{x}_t) - \Delta t\, \nabla_{\mathbf{x}} \cdot f(\mathbf{x}_t, t) + O(\Delta t^2).\end{aligned}$$

And therefore,

$$\log p_{t+\Delta t}(x_{t+\Delta t}) - \log p_t(x_t) = -\Delta t\, \nabla_x \cdot f(x_t, t) + O(\Delta t^2). \quad \text{(B.1.5)}$$

Let us now starts from the first order Taylor expansion of the LHS,

$$\begin{aligned}&\log p_{t+\Delta t}(\mathbf{x}_{t+\Delta t}) - \log p_t(\mathbf{x}_t) \\ &\quad = \Delta t\, \partial_t \log p_t(x_t) + (\mathbf{x}_{t+\Delta t} - \mathbf{x}_t)^\top \nabla_x \log p_t(\mathbf{x}_t) + O(\Delta t^2) \\ &\quad = \Delta t\, \partial_t \log p_t(\mathbf{x}_t) + \Delta t\, f(\mathbf{x}_t, t)^\top \nabla_{\mathbf{x}} \log p_t(\mathbf{x}_t) + O(\Delta t^2).\end{aligned}$$

Matching this with the RHS of (B.1.5) and letting $\Delta t \to 0$ yields

Derived Distribution of an ODE (3)

$$\partial_t \log p_t(\mathbf{x}_t) = -\nabla_{\mathbf{x}} \cdot f(\mathbf{x}_t, t) - f(\mathbf{x}_t, t)^\top \nabla_{\mathbf{x}} \log p_t(\mathbf{x}_t). \tag{10}$$

lifting log by chain rule

$$\begin{aligned}
\partial_t p_t(\mathbf{x}_t) &= p_t(\mathbf{x}_t) \partial_t \log p_t(\mathbf{x}_t) \\
&= p_t(\mathbf{x}_t)[-\nabla_{\mathbf{x}} \cdot f(\mathbf{x}_t, t) - f(\mathbf{x}_t, t)^\top \nabla_{\mathbf{x}} \log p_t(\mathbf{x}_t)] \\
&= [-p_t(\mathbf{x}_t) \nabla_{\mathbf{x}} \cdot f(\mathbf{x}_t, t) - f(\mathbf{x}_t, t)^\top p_t(\mathbf{x}_t) \nabla_{\mathbf{x}} \log p_t(\mathbf{x}_t)] \\
&= [-p_t(\mathbf{x}_t) \nabla_{\mathbf{x}} \cdot f(\mathbf{x}_t, t) - f(\mathbf{x}_t, t)^\top \nabla_{\mathbf{x}} \cdot p_t(\mathbf{x}_t)] \\
&= -\nabla_{\mathbf{x}} \cdot (p_t(\mathbf{x}_t) f(\mathbf{x}_t, t))
\end{aligned}$$

Proof of Fokker–Planck Equation

The proof of the FP equation includes the following

- Second-order expansion of an evaluation function w.r.t a state variable following an ODE.
- Replacing the state variable with respect to an SDE.
- Taking the mean of the chage of the evaluation function.

Chain Rules from Taylor Expansion

Let us start from the chain rule for a deterministic process.

Consider x_t following some ODE $dx_t = f(x,t)dt$. For a nice function $h(x_t, t)$ if we want to inspect its evolution w.r.t. time,

$$dh = \left(\frac{\partial h}{\partial t} + \frac{\partial h}{\partial x_t} f\right) dt$$

We write the Taylor expansion up to second order term,

$$h(x_{t+dt},\, t+dt) - h(x_t,\, t) = \frac{\partial h}{\partial t}\, dt + \frac{\partial h}{\partial x}\, dx_t + \frac{1}{2}\left(\frac{\partial^2 h}{\partial x^2}(dx_t)^2 + 2\,\frac{\partial^2 h}{\partial t \partial x}\, dt\, dx_t + \frac{\partial^2 h}{\partial t^2}\,(dt)^2\right)$$

Stochastic Chain Rules (Ito's Lemma) (1)

If our x_t now actually follows an SDE,

$$dx_t = f(x_t, t)\,dt + g(t)\,dw_t$$

We still can have the same Taylor expansion, but we need to inspect the cross terms carefully since they don't easily vanishes w.r.t. dt.

- **Negligible Cross Terms.**
  By Brownian motion,
  - $dw_t = O(\sqrt{dt}) \quad \Longrightarrow \quad dt \cdot dw_t = O\big((dt)^{3/2}\big)$.
  - $dt\,dx_t = dt\,(f\,dt + g\,dw_t) = f(dt)^2 + g\,dt \cdot dw_t = O\big((dt)^{3/2}\big)$.

Stochastic Chain Rules (Ito's Lemma) (2)

- **Second-Order Term $(dx_t)^2$.** Expanding using the SDE,

$$\begin{aligned}(dx_t)^2 &= (f\,dt + g\,dw_t)^2 \\ &= f^2(dt)^2 + 2fg\,dt\,dw_t + g^2(dw_t)^2 \\ &= O\big((dt)^2\big) + O\Big((dt)^{3/2}\Big) + g^2\,O(dt) \\ &= g^2(t)\,dt + O\Big((dt)^{3/2}\Big).\end{aligned}$$

- We therefore obtain the differential,

$$dh(x_t, t) = \frac{\partial h}{\partial t}\,dt + \frac{\partial h}{\partial x}\,dx_t + \frac{1}{2}\,\frac{\partial^2 h}{\partial x^2}\,g^2(t)\,dt.$$

Substituting

$$dx_t = f(x_t, t)\,dt + g(t)\,dw_t,$$

yields

Stochastic Chain Rules (Ito's Lemma) (3)

$$dh(x_t, t) = \left(\frac{\partial h}{\partial t} + f \frac{\partial h}{\partial x} + \frac{1}{2} g^2 \frac{\partial^2 h}{\partial x^2} \right) dt \; + \; g \frac{\partial h}{\partial x} dw_t.$$

This is the 1D Itô's formula. The multivariate case follows immediately.

Stochastic Chain Rules (Ito's Lemma) (4)

Multidimensional Itô Formula

Let $\mathbf{x}_t \in \mathbb{R}^D$ satisfy the SDE

$$d\mathbf{x}_t = f(\mathbf{x}_t, t)\, dt + g(t)\, d\mathbf{w}_t,$$

where $f : \mathbb{R}^n \times [0, T] \to \mathbb{R}^n$, $g : [0, T] \to \mathbb{R}$, and $\mathbf{w}_t \in \mathbb{R}^n$ is an n-dimensional Brownian motion.
Let $h : \mathbb{R}^n \times [0, T] \to \mathbb{R}$ be a scalar-valued function. Then $h(\mathbf{x}_t, t)$ satisfies:

$$dh(\mathbf{x}_t, t) = \left(\frac{\partial h}{\partial t} + \nabla_{\mathbf{x}} h^\top f + \frac{1}{2} g^2(t) \operatorname{Tr}\left(\nabla_{\mathbf{x}}^2 h\right) \right) dt \; + \; g(t)\, \nabla_{\mathbf{x}} h^\top d\mathbf{w}_t.$$

Deriving the Fokker–Planck Equation (1)

Let $\phi(x,t)$ be a smooth test function $\phi : \mathbb{R}^n \times [0,T] \to \mathbb{R}$ with compact support in time.

By Itô's formula, we know the derivative

$$d\phi(x_t,t) = \left(\frac{\partial \phi}{\partial t} + \nabla_x \phi^\top f(x_t,t) + \frac{1}{2} g^2(t) \operatorname{Tr}\left[\nabla_x^2 \phi\right] \right) dt + g(t)\, \nabla_x \phi^\top dw_t.$$

Taking expectation over the law p_t and using $\mathbb{E}[dw_t] = 0$,

$$\begin{aligned} \mathbb{E}[d\phi(x_t,t)] &= \mathbb{E}\left[\left(\frac{\partial \phi}{\partial t} + \nabla_x \phi^\top f(x_t,t) + \frac{1}{2} g^2(t) \operatorname{Tr}(\nabla_x^2 \phi) \right) dt \right]. \\ &= \int \left(\frac{\partial \phi}{\partial t} + \nabla_x \phi^\top f(x,t) + \frac{1}{2} g^2(t) \operatorname{Tr}(\nabla_x^2 \phi) \right) p_t(x)\, dx\, dt. \end{aligned}$$

Deriving the Fokker–Planck Equation (2)

If we dare to recall our Vector Calculus, using divergence theorem,

$$\int \nabla_x \phi^\top f \, p_t \, dx = -\int \phi \, \nabla_x \cdot (f p_t) \, dx,$$

$$\int \mathrm{Tr}(\nabla_x^2 \phi) \, p_t \, dx = -\int \phi \, \Delta p_t \, dx.$$

Plugging back to the expectation and we obtain,

$$\mathbb{E}[d\phi(x_t, t)] = \int \phi(x, t) \left[\frac{\partial p_t}{\partial t} + \nabla_x \cdot (f p_t) - \frac{1}{2} g^2(t) \, \Delta p_t \right] dx \, dt.$$

Since ϕ is arbitrary, the integrand must vanish. Setting the terms inside the parenthesis and we arrived at,

$$\begin{aligned} \frac{\partial p_t}{\partial t} &= -\nabla_{\mathbf{x}} \cdot (f(\mathbf{x}, t) \, p_t(\mathbf{x})) + \frac{1}{2} g^2(t) \, \Delta p_t(\mathbf{x}) \\ &= -\nabla_{\mathbf{x}} \cdot (f - \frac{1}{2} g^2(t) \nabla_{\mathbf{x}} \log p_t(\mathbf{x})) \end{aligned}$$